THE COMPLETION MAP OF A G-RING IS REGULAR

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness and Noetherianity of completions. Algebra, Lemma \ref{algebra-lemma-completion-flat} and Algebra, Lemma \ref{algebra-lemma-completion-Noetherian-Noetherian}, cited at proof lines 12704-12707. Allow the local criterion for regular ring maps to be applied.
- The local criterion for a regular ring map. Lemma \ref{lemma-regular-local}, cited at proof lines 12708-12709. Reduces regularity to the behaviour at maximal ideals of the target.
- The ideal of definition is contained in every maximal ideal of the completion and the quotient of the completion by it is the quotient of the ring. Algebra, Lemma \ref{algebra-lemma-radical-completion}, cited at proof line 12712. Identify residue fields and show that the prime below is maximal.
- A flat local map of Noetherian local rings identifying residue fields and with equal maximal ideals induces an isomorphism on completions. Lemma \ref{lemma-flat-unramified}, cited at proof lines 12722-12723. Identifies the completion of the localised completion with the completion of the localisation.
- Permanence of regularity for a composition where one map is faithfully flat, and regularity of the completion of a G-ring at a prime. Lemma \ref{lemma-regular-permanence}, cited at proof lines 12729-12730, and Algebra, Lemma \ref{algebra-lemma-completion-faithfully-flat}, cited at proof lines 12726-12728. Apply the permanence statement to the two-step chain, the outer map being regular because the ring is a G-ring.

Source reference. Original lemma 0AH2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13215–13215. The lemma's own canonical tag,

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quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0316: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3973–3973. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07C0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9360–9360. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7062–7062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AGX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13210–13210. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00MC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 818–818. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9773–9773. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-MatCA: sources/stacks-more-algebra/pinned-original-source.tex, source lines 12693–12693. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let A be a G-ring. Let $I \subset A$ be an ideal and let $A^\wedge$ be the completion of A with respect to I. Then $A \to A^\wedge$ is regular.*

Proof. The ring map $A \to A^\wedge$ is flat by Algebra, Lemma 00MB. The ring $A^\wedge$ is Noetherian by Algebra, Lemma 0316. Thus it suffices to check the third condition of Lemma 07C0. Let $\mathfrak{m}' \subset A^\wedge$ be a maximal ideal lying over $\mathfrak{m} \subset A$. By Algebra, Lemma 05GI we have $IA^\wedge \subset \mathfrak{m}'$. Since $A^\wedge/IA^\wedge = A/I$ we see that $I \subset \mathfrak{m}$, $\mathfrak{m}/I = \mathfrak{m}'/IA^\wedge$, and $A/\mathfrak{m} = A^\wedge/\mathfrak{m}'$. Since $A^\wedge/\mathfrak{m}'$ is a field, we conclude that $\mathfrak{m}$ is a maximal ideal as well. Then $A_\mathfrak{m} \to A^\wedge_{\mathfrak{m}'}$ is a flat local ring homomorphism of Noetherian local rings which identifies residue fields and such that $\mathfrak{m}A^\wedge_{\mathfrak{m}'} = \mathfrak{m}'A^\wedge_{\mathfrak{m}'}$. Thus it induces an isomorphism on complete local rings, see Lemma 0AGX. Let $(A_\mathfrak{m})^\wedge$ be the completion of $A_\mathfrak{m}$ with respect to its maximal ideal. The ring map

$$(A^\wedge)_{\mathfrak{m}'} \to ((A^\wedge)_{\mathfrak{m}'})^\wedge = (A_\mathfrak{m})^\wedge$$

is faithfully flat (Algebra, Lemma 00MC). Thus we can apply Lemma 07NT to the ring maps

$$A_\mathfrak{m} \to (A^\wedge)_{\mathfrak{m}'} \to (A_\mathfrak{m})^\wedge$$

to conclude because $A_{\mathfrak{m}} \to (A_{\mathfrak{m}})^{\wedge}$ is regular as A is a G-ring. $\square$